

Prophylactic Radiation Therapy Versus Standard of Care for Patients With High-Risk Asymptomatic Bone Metastases: A Multicenter, Randomized Phase II Clinical Trial

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Publication: Journal of Clinical Oncology Volume 42, Number 1 <https://doi.org/10.1200/JCO.23.00753>

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Abstract

Purpose

External-beam radiation therapy (RT) is standard of care (SOC) for pain relief of symptomatic bone metastases. We aimed to evaluate the efficacy of radiation to asymptomatic bone metastases in preventing skeletal-related events (SRE).

Methods

In a multicenter randomized controlled trial, adult patients with widely metastatic solid tumor malignancies were stratified by histology and planned SOC (systemic therapy or observation) and randomly assigned in a 1:1 ratio to receive RT to asymptomatic high-risk bone metastases or SOC alone. The primary outcome of the trial was SRE. Secondary outcomes included hospitalizations for SRE and overall survival (OS).

Results

A total of 78 patients with 122 high-risk bone metastases were enrolled between May 8, 2018, and August 9, 2021, at three institutions across an affiliated cancer network in the United States. Seventy-three patients were evaluable for the primary end point. The most common primary cancer types were lung (27%), breast (24%), and prostate (22%). At 1 year, SRE occurred in one of 62 bone metastases (1.6%) in the RT arm and 14 of 49 bone metastases (29%) in the SOC arm ($P < .001$). There were significantly fewer patients hospitalized for SRE in the RT arm compared with the SOC arm (0 v 4 , $P = .045$). At a median follow-up of 2.5 years, OS was significantly longer in the RT arm (hazard ratio [HR], 0.49; 95% CI, 0.27 to 0.89; $P = .018$), which persisted on multivariable Cox regression analysis (HR, 0.46; 95% CI, 0.23 to 0.85; $P = .01$).

Conclusion

Radiation delivered prophylactically to asymptomatic, high-risk bone metastases reduced SRE and hospitalizations. We also observed an improvement in OS with prophylactic radiation, although a confirmatory phase III trial is warranted.



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Study Protocol

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Please click the protocol link below to access the information.

- [Protocol](#)

Prior Presentation

Presented in part at the American Society of Radiation Oncology Annual Meeting, San Antonio, TX, October 23, 2022.

Support

Supported in part by the National Institutes of Health/National Cancer Institute (NIH/NCI) through Cancer Center Support Grant No. P30 CA008748 (Memorial Sloan Kettering) and NCI Career Development Award No. K08 CA252640 (E.F.G.).

Clinical Trial Information

[NCT03523351](#)

Authors' Disclosures of Potential Conflicts of Interest

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Research Funding: Elekta (Inst), Amgen (Inst), Merck (Inst), Regeneron (Inst), EMD Serono/Merck (Inst), PSI (Inst), Alpha TAU Medical (Inst)

Uncompensated Relationships: Castle Biosciences

Open Payments Link: <https://openpaymentsdata.cms.gov/physician/437311>

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Speakers' Bureau: Novocure, GT Medical Technologies

Research Funding: Medtronic (Inst), Blue Earth Diagnostics (Inst), Novocure (Inst), GT Medical Technologies (Inst), AstraZeneca (Inst), Exelixis (Inst), ViewRay (Inst), BrainLAB (Inst), Cantex Pharmaceuticals, Inc (Inst)

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Patents, Royalties, Other Intellectual Property: Daniel Higginson and Rama Damerla are inventors on a provisional patent application filed by their institution, Memorial Sloan Kettering Cancer Center (Inst), Daniel Higginson and Jonathan Leeman are inventors on a provisional patent application filed by their institution, Memorial Sloan Kettering Cancer Center (Inst), Daniel Higginson and Suleman Hussain are inventors on a provisional patent application filed by their institution, Memorial Sloan Kettering Cancer Center (Inst), Daniel Higginson and Grace Moore are inventors on a provisional patent application filed by their institution, Memorial Sloan Kettering Cancer Center (Inst)

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Stock and Other Ownership Interests: Nanocan Therapeutics

Consulting or Advisory Role: Galera Therapeutics, Nanocan Therapeutics, Kazia Therapeutics, AstraZeneca, Plus Therapeutics

Research Funding: AstraZeneca, Kazia Therapeutics, X-RAD Therapeutics, Debiopharm Group, Biocept

No other potential conflicts of interest were reported.

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